# Questions

Q1.

The equation

$$\frac{3}{x} + 5 = -2x + c$$

where $c$ is a constant, has no real roots.

Find the range of possible values of $c$.

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**(Total for question = 7 marks)**

**(Q09 WMA11/01, Jan 2019)**

Q2.

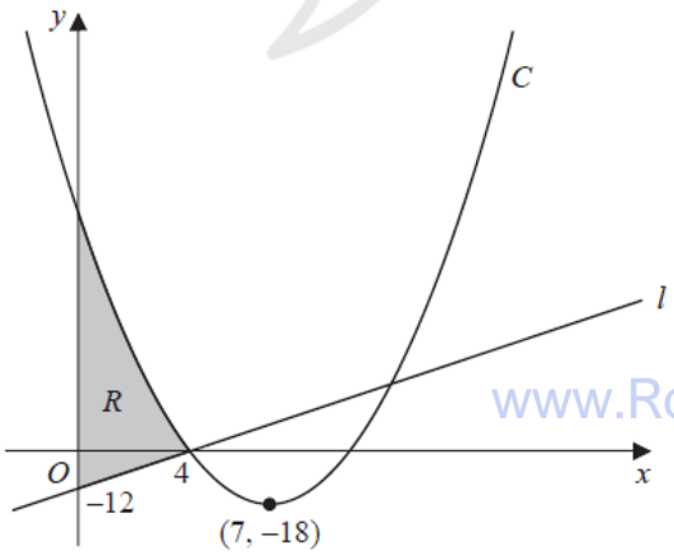



**Figure 2**

Figure 2 shows a sketch of the straight line $l$ and the curve $C$.

Given that $l$ cuts the $y$-axis at –12 and cuts the $x$-axis at 4, as shown in Figure 2,

(a) find an equation for l, writing your answer in the form $y = mx + c$, where $m$ and $c$ are constants to be found.

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**(2)**

Given that $C$

- has equation $y = f(x)$ where $f(x)$ is a quadratic expression
- has a minimum point at $(7, -18)$
- cuts the $x$-axis at 4 and at $k$, where $k$ is a constant

(b) deduce the value of $k$,

**(1)**

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(c) find $f(x)$.

**(3)**

The region $R$ is shown shaded in Figure 2.

(d) Use inequalities to define $R$.

**(2)**

**(Total for question = 8 marks)**

(**Q08 WMA11/01, Jan 2023**)

Q3.

Given that the equation

$$kx^2 + 6kx + 5 = 0 \text{ where } k \text{ is a non zero constant}$$

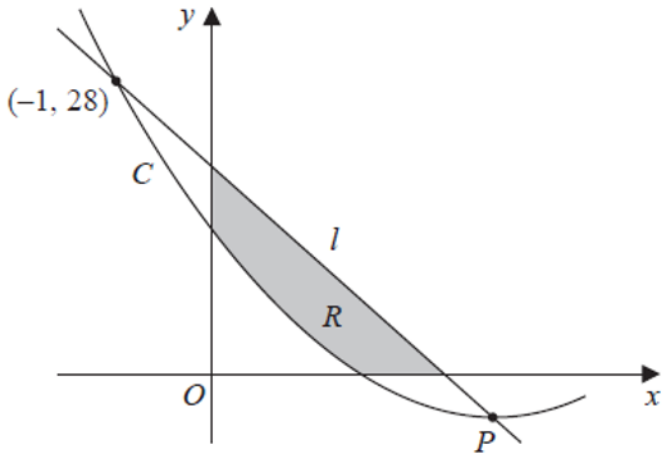
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has no real roots, find the range of possible values for $k$.

**(Total for question = 4 marks)**

(**Q04 WMA11/01, Jan 2023**)

Q4.




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**Figure 5**

Figure 5 shows part of the curve $C$ with equation $y = f(x)$ where

$$f(x) = 2x^2 - 12x + 14$$

(a) Write $2x^2 - 12x + 14$ in the form

$$a(x + b)^2 + c$$

where $a$, $b$ and $c$ are constants to be found.

**(3)**

Given that $C$ has a minimum at the point $P$

(b) state the coordinates of $P$

**(1)**

The line $l$ intersects $C$ at $(-1, 28)$ and at $P$ as shown in Figure 5.

(c) Find the equation of $l$ giving your answer in the form $y = mx + c$ where $m$ and $c$ are constants to be found.

**(3)**

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The finite region $R$, shown shaded in Figure 5, is bounded by the $x$-axis, $l$, the $y$-axis, and $C$.

(d) Use inequalities to define the region $R$.

**(3)**

**(Total for question = 10 marks)**

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Q5.

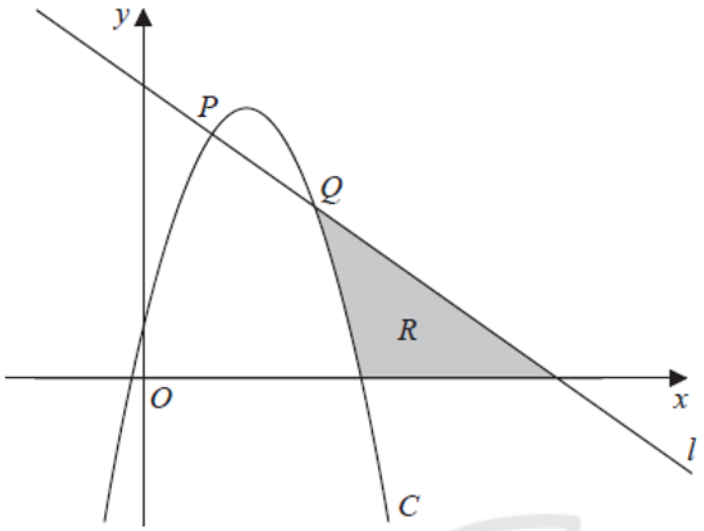



**Figure 1**

**In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

Figure 1 shows a line $l$ with equation $x + y = 6$ and a curve $C$ with equation $y = 6x - 2x^2 + 1$

The line $l$ intersects the curve $C$ at the points $P$ and $Q$ as shown in Figure 1.

(a) Find, using algebra, the coordinates of $P$ and the coordinates of $Q$.

**(4)**

The region $R$, shown shaded in Figure 1, is bounded by $C$, $l$ and the $x$-axis.

(b) Use inequalities to define the region $R$.

**(3)**

**(Total for question = 7 marks)**



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Q6.

(i) Solve

$$\frac{3}{x} > 4$$

**(3)**

(ii)

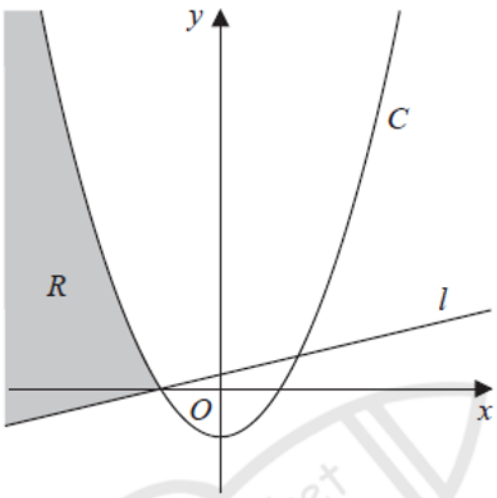



**Figure 1**

Figure 1 shows a sketch of the curve $C$ and the straight line $l$.

The infinite region $R$, shown shaded in Figure 1, lies in quadrants 2 and 3 and is bounded by $C$ and $l$ only.

Given that

- $l$ has a gradient of 3
- $C$ has equation $y = 2x^2 - 50$
- $C$ and $l$ intersect on the negative $x$-axis

use inequalities to define the region $R$.

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**(3)**

**(Total for question = 6 marks)**

**(Q03 WMA11/01, Oct 2021)**

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Q7.

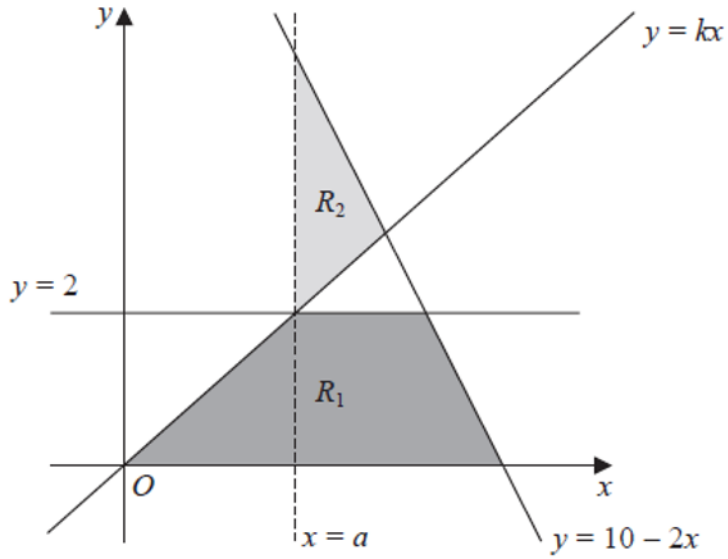



**Figure 2**

The region $R_1$, shown shaded in Figure 2, is defined by the inequalities

$$0 \leq y \leq 2 \qquad y \leq 10 - 2x \qquad y \leq kx$$

where $k$ is a constant.

The line $x = a$, where $a$ is a constant, passes through the intersection of the lines $y = 2$ and $y = kx$

Given that the area of $R_1$ is $\frac{27}{4}$ square units,

(a) find

(i) the value of $a$

(ii) the value of $k$

**(4)**

(b) Define the region $R_2$, also shown shaded in Figure 2, using inequalities.

**(2)**

**(Total for question = 6 marks)**

**(Q07 WMA11/01, June 2023)**

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Q8.

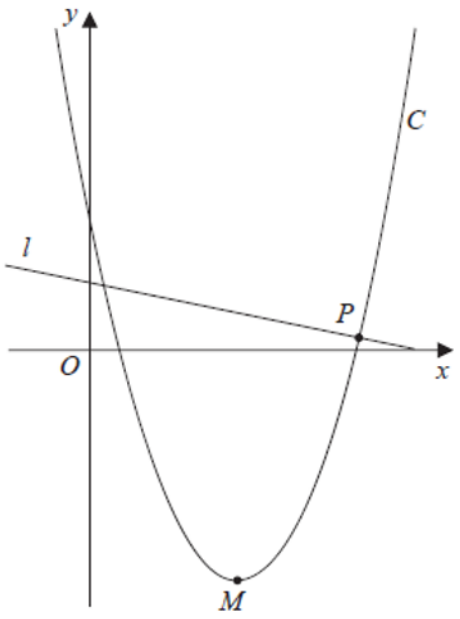



**Figure 3**

Figure 3 shows a sketch of the curve $C$ with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

(a) Write $\frac{1}{2}x^2 - 10x + 22$ in the form

$$a(x + b)^2 + c$$

where $a$, $b$ and $c$ are constants to be found.

**(3)**

The point $M$ is the minimum turning point of $C$, as shown in Figure 3.

(b) Deduce the coordinates of $M$

**(2)**

The line $l$ is the normal to $C$ at the point $P$, as shown in Figure 3.

Given that $l$ has equation $y = k - \frac{1}{8}x$, where $k$ is a constant,

(c) (i) find the coordinates of $P$

(ii) find the value of $k$

**(6)**

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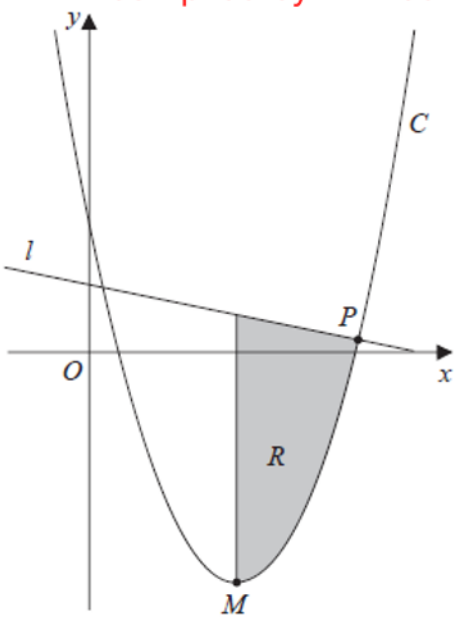



**Figure 4**

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Figure 4 is a copy of Figure 3. The finite region $R$, shown shaded in Figure 4, is bounded by $l$, $C$ and the line through $M$ parallel to the $y$-axis.

(d) Identify the inequalities that define $R$.

**(3)**

**(Total for question = 14 marks)**

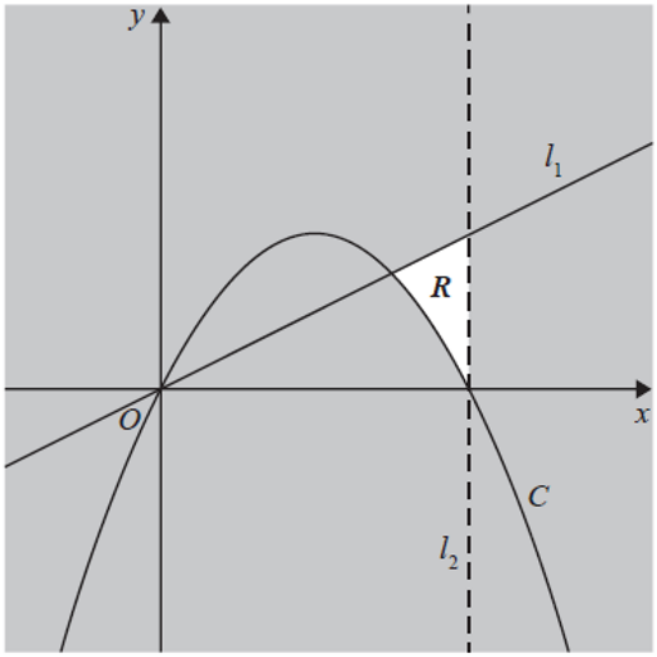
(**Q09 WMA11/01, Oct 2022**)

Q9.

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**Figure 1**

Figure 1 shows a line $l_1$ with equation $2y = x$ and a curve $C$ with equation $y = 2x - \frac{1}{8}x^2$

The region $R$, shown unshaded in Figure 1, is bounded by the line $l_1$, the curve $C$ and a line $l_2$

Given that $l_2$ is parallel to the $y$-axis and passes through the intercept of $C$ with the positive $x$-axis, identify the inequalities that define $R$.

**(Total for question = 3 marks)**

**(Q04 WMA11/01, Jan 2019)**

Q10.

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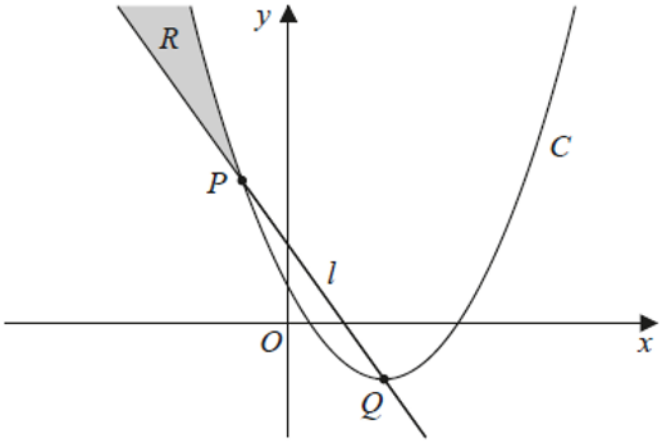



**Figure 2**

The points $P$ and $Q$, as shown in Figure 2, have coordinates (–2, 13) and (4, –5) respectively.

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The straight line $l$ passes through $P$ and $Q$.

(a) Find an equation for $l$, writing your answer in the form $y = mx + c$, where $m$ and $c$ are integers to be found.

**(3)**

The quadratic curve $C$ passes through $P$ and has a minimum point at $Q$.

(b) Find an equation for $C$.

**(3)**

The region $R$, shown shaded in Figure 2, lies in the second quadrant and is bounded by $C$ and $l$ only.

(c) Use inequalities to define region $R$.

**(2)**

**(Total for question = 8 marks)**

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**(Q04 WMA11/01, Jan 2021)**

Q11.

**In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

Solve the inequality

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$4x^2 - 3x + 7 \geq 4x + 9$

**(Total for question = 4 marks)**

**(Q01 WMA11/01, June 2023)**

Q12.

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The curve $C$ has equation $y = f(x)$

Given that

- $f(x)$ is a quadratic expression
- the maximum turning point on $C$ has coordinates $(-2, 12)$
- $C$ cuts the negative $x$-axis at $-5$

(a) find $f(x)$

**(4)**

The line $l_1$ has equation $y = \frac{4}{5}x$

Given that the line $l_2$ is perpendicular to $l_1$ and passes through $(-5, 0)$

(b) find an equation for $l_2$ , writing your answer in the form $y = mx + c$ where $m$ and $c$ are constants to be found.

**(3)**

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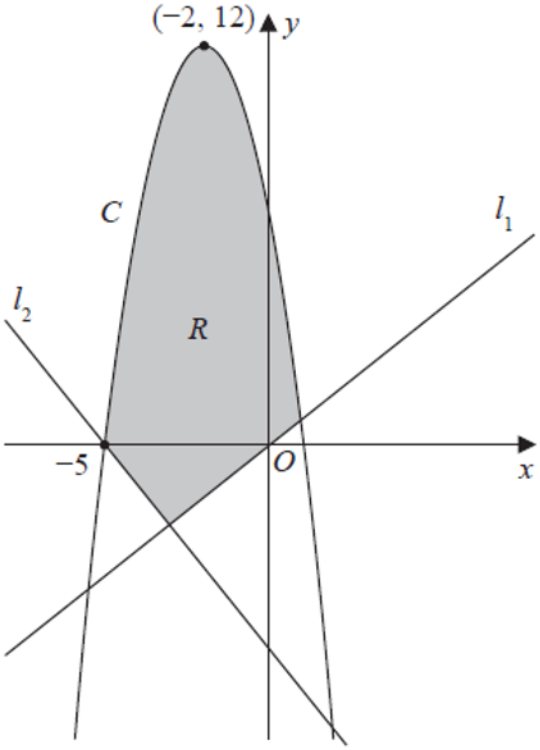



**Figure 2**

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Figure 2 shows a sketch of the curve $C$ and the lines $l_1$ and $l_2$

(c) Define the region $R$, shown shaded in Figure 2, using inequalities.

**(2)**

**(Total for question = 9 marks)**

(**Q05 WMA11/01, June 2022**)

Q13.

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(a) On the same axes, sketch the graphs of $y = x + 2$ and $y = x^2 - x - 6$ showing the coordinates of all points at which each graph crosses the coordinate axes.

**(4)**

(b) On your sketch, show, by shading, the region $R$ defined by the inequalities

$$y < x + 2 \textbf{ and } y > x^2 - x - 6$$

**(1)**

(c) Hence, or otherwise, find the set of values of $x$ for which $x^2 - 2x - 8 < 0$

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**(3)**

**(Total for question = 8 marks)**

**(Q05 WMA11/01, Specimen papers)**

Q14.

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The straight line $l$ has equation $y = k(2x - 1)$, where $k$ is a constant.

The curve $C$ has equation $y = x^2 + 2x + 11$

Find the set of values of $k$ for which $l$ does not cross or touch $C$.

**(6)**

**(Total for question = 6 marks)**

**(Q08 WMA11/01, Jan 2020)**

Q15.

**In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

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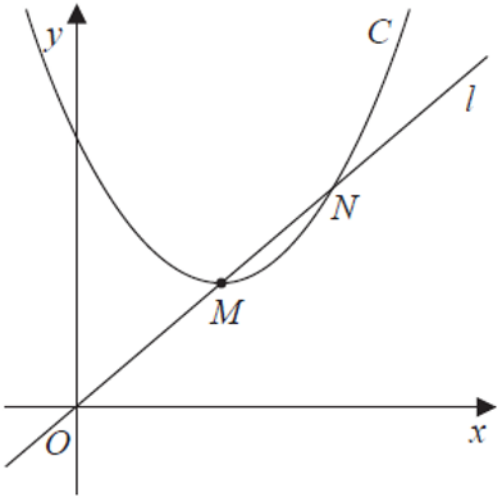



**Figure 2**

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Figure 2 shows a sketch of the curve $C$ with equation $y = x^2 - 5x + 13$

The point $M$ is the minimum point of $C$.

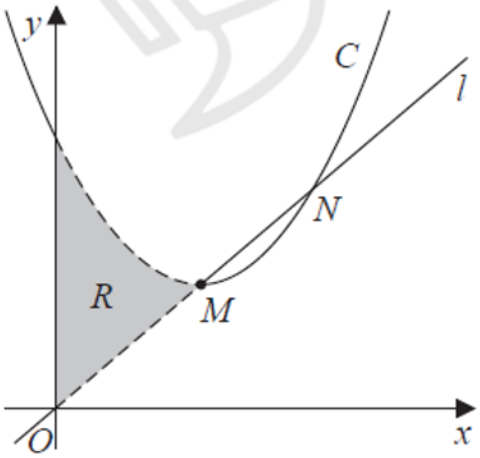
The straight line $l$ passes through the origin $O$ and intersects $C$ at the points $M$ and $N$ as shown.

Find, showing your working,

(a) the coordinates of $M$,

**(3)**

(b) the coordinates of $N$.




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**Figure 3**

Figure 3 shows the curve $C$ and the line $l$. The finite region $R$, shown shaded in Figure 3, is bounded by $C$, $l$ and the $y$-axis.

(c) Use inequalities to define the region $R$.

**(2)**

**(Total for question = 5 marks)**

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(**Q03 WMA11/01, Oct 2019**)

Q16.

The straight line with equation $y = 4x + c$, where $c$ is a constant, is a tangent to the curve with equation $y = 2x^2 + 8x + 3$

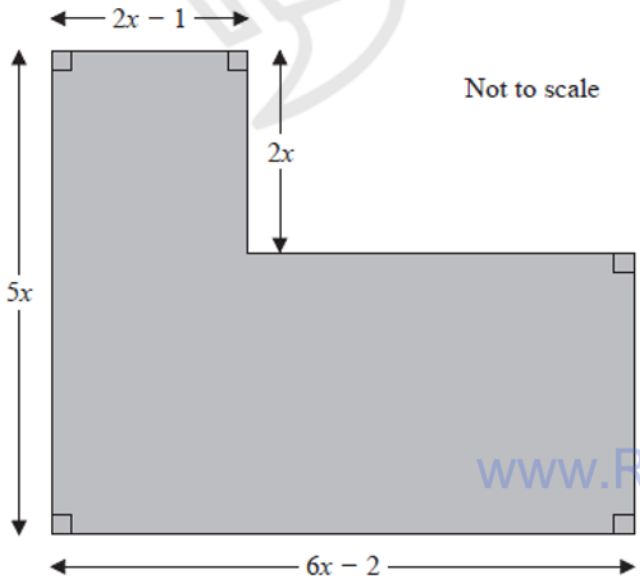
Calculate the value of $c$

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**(Total for question = 5 marks)**

(**Q04 WMA11/01, Specimen papers**)

Q17.




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**Figure 1**

Figure 1 shows the plan of a garden. The marked angles are right angles.

The six edges are straight lines.

The lengths shown in the diagram are given in metres.

Given that the perimeter of the garden is greater than 29 m,

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(a) show that $x > 1.5$ m

**(3)**

Given also that the area of the garden is less than 72 $m^2$,

(b) form and solve a quadratic inequality in $x$.

**(5)**

(c) Hence state the range of possible values of $x$.

**(1)**

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**(Total for question = 9 marks)**

**(Q03 WMA11/01, June 2019)**

Q18.

The curve $C$ has equation

$$y = \frac{1}{x^2} - 9$$

(a) Sketch the graph of $C$.

On your sketch

- show the coordinates of any points of intersection with the coordinate axes
- state clearly the equations of any asymptotes

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**(4)**

The curve $D$ has equation $y = kx^2$ where $k$ is a constant.

Given that $C$ meets $D$ at 4 distinct points,

(b) find the range of possible values for $k$.

**(5)**

**(Total for question = 9 marks)**

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**(Q010 WMA11/01, Jan 2022)**

Q19.

**In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

The equation

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$$4(p-2x)=\frac{12+15p}{x+p} \qquad x \neq -p$$

where $p$ is a constant, has two distinct real roots.

(a) Show that

$$3p^2-10p-8>0$$

**(3)**

(b) Hence, using algebra, find the range of possible values of $p$

**(3)**

**(Total for question = 6 marks)**

**(Q06 WMA11/01, Oct 2023)**

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Q20.

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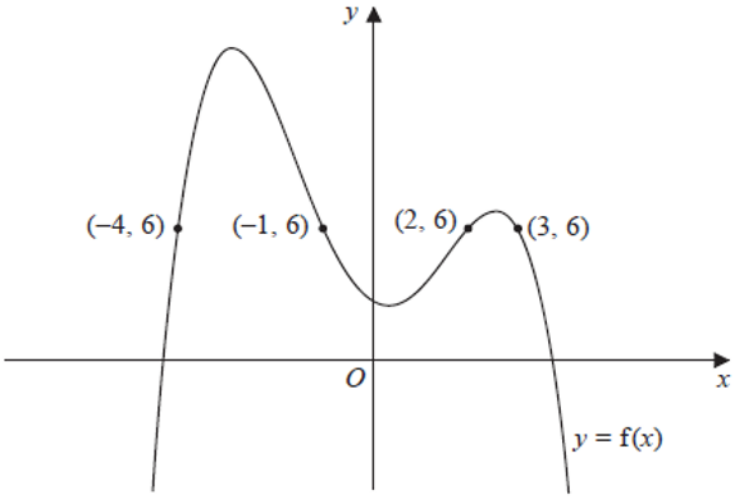



**Figure 1**

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Figure 1 shows the curve with equation $y = f(x)$.

The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

(a) Using Figure 1, find the range of values of $x$ for which

$$f(x) < 6$$

**(3)**

(b) State the largest solution of the equation

$$f(2x) = 6$$

**(1)**

(c) (i) Sketch the curve with equation $y = f(-x)$.

On your sketch, state the coordinates of the points to which $P$, $Q$, $R$ and $S$ are transformed.

(ii) Hence find the set of values of $x$ for which

$$f(-x) \geqslant 6 \text{ and } x < 0$$

**(4)**

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**(Total for question = 8 marks)**

**(Q07 WMA11/01, Oct 2022)**

Q21.

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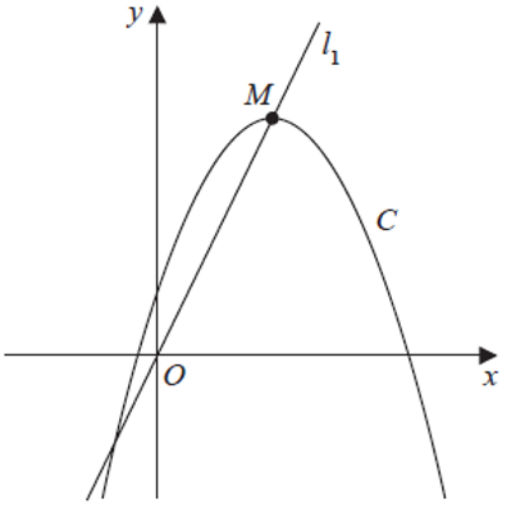



**Figure 4**

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Figure 4 shows a sketch of the curve $C$ with equation

$$y = 4 + 12x - 3x^2$$

The point $M$ is the maximum turning point on $C$.

(a) (i) Write $4 + 12x - 3x^2$ in the form

$$a + b(x + c)^2$$

where $a$, $b$ and $c$ are constants to be found.

(ii) Hence, or otherwise, state the coordinates of $M$.

**(5)**

The line $l_1$ passes through $O$ and $M$, as shown in Figure 4.

A line $l_2$ touches $C$ and is parallel to $l_1$

(b) Find an equation for $l_2$

**(5)**

**(Total for question = 10 marks)**

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